PHYS 120 · A SHAPED BOARD WITH DESIGN CALCULATIONS, DEFENDED

Design calculations

Rowan Hale · defended 1 December 2027 · shaped in WKSP 102

1. Volume. Thirteen cross-sections at 6-inch stations, each area from the template's width and the foil's thickness, summed by Simpson's rule: $V \approx (h/3)[A_0 + 4A_1 + 2A_2 + \dots + A_{12}] = 38.4$ L.

2. Buoyancy. 38.4 L of foam, glass and resin displaces 38.4 kg of seawater ($\rho = 1{,}025$ kg/m³ at 14 °C) and weighs 3.1 kg, so the board holds 35.3 kg before I touch it. A 72 kg rider sits the board 2.0 L under at rest, which is the paddling depth I measured on the old board and wanted to keep.

3. Fins. The Cove's waves in PHYS 110 ran 4.1–6.3 m/s on the face. A twin keel of 5¼" base and 4½" depth gives about 95 cm² a side; at 5 m/s and 8° of yaw that is roughly 110 N of lateral force a side, enough to hold a line on a waist-high wall and let go on a cutback, which is what a fish is for.

4. Rocker. 4¼" nose, 1⅞" tail, a continuous curve so the board paddles flat and still turns. Less rocker than the old board by ¾", which is why it paddles into waves a second earlier.

Defended before the PHYS 120 panel with the board on sawhorses. Asked: what would you change? More tail rocker for the winter. The numbers say so and now so does the riding.

The stations

station	x (in)	width	thick	area cm²	weight	contrib L
0	0	3.0	0.60	8.4	1	0.04
1	6	9.5	1.20	53.0	4	1.08
2	12	14.0	1.70	110.6	2	1.12
3	18	17.2	2.10	167.8	4	3.41
4	24	19.4	2.40	216.3	2	2.20
5	30	20.6	2.55	244.0	4	4.96
6	36	21.0	2.62	256.1	2	2.60
7	42	20.7	2.55	245.2	4	4.98
8	48	19.6	2.35	214.0	2	2.17
9	54	17.8	2.00	165.4	4	3.36
10	60	15.5	1.60	115.2	2	1.17
11	66	13.0	1.20	72.5	4	1.47
12	72	10.5	0.80	39.0	1	0.20

ROCKER PROFILE AND VOLUME BY STATION



